

# Introduction to Python

## Module 5 — Strings & Files

String methods, splitting, and file I/O

# Strings Are Powerful

Text is central to a lot of research, and Python strings come with dozens of useful **methods**. A method is a function attached to a value, called with a dot.

```
text = " Hello World "  
  
text.lower()      # " hello world "  
text.upper()      # " HELLO WORLD "  
text.strip()      # "Hello World" (trim spaces)  
text.replace("World", "Python")
```

*These are exactly the operations behind text preprocessing — lowercasing, trimming, cleaning.*

## Splitting & Joining

Two workhorse methods for turning text into lists and back — the heart of tokenization.

```
sentence = "the cat sat"

# split a string into a list of words
words = sentence.split() # ["the", "cat", "sat"]

# join a list back into a string
" ".join(words)          # "the cat sat"
"-".join(words)          # "the-cat-sat"
```

`split()` on nothing splits on spaces; `split(",")` splits on commas (handy for CSV lines).  
`join()` glues a list together with the string you call it on.

## Checking & Finding in Strings

```
word = "policy"

"pol" in word      # True (substring check)
word.startswith("pol") # True
word.endswith("cy")  # True
word.count("o")      # 1
len(word)           # 6
```

The `in` keyword works on strings (“is this substring present?”) just like it works on lists. These checks are the building blocks of filtering and searching text.

## Reading & Writing Files

Real data lives in files. The safe way to open one is with a `with` block — it closes the file automatically when you're done.

```
# writing
with open("notes.txt", "w") as f:
    f.write("Hello file")

# reading
with open("notes.txt") as f:
    content = f.read()
```

The mode `"w"` **writes** (overwrites!), `"r"` **reads** (the default), `"a"` **appends**. For structured data (CSVs, spreadsheets), you'll use **pandas** — coming next.

## Module 5: Takeaways

- ✓ String **methods**: `.lower()` `.upper()` `.strip()` `.replace()`
- ✓ `.split()` makes a list of words; `.join()` glues them back
- ✓ `in`, `.startswith()`, `.count()` check and search text
- ✓ Open files safely with `with open(...) as f:`
- ✓ Modes: "r" read, "w" write, "a" append

**Next:** Module 6 — **NumPy**, the foundation of numerical computing in Python.